

Probability Theory and Statistics

Lecture 14

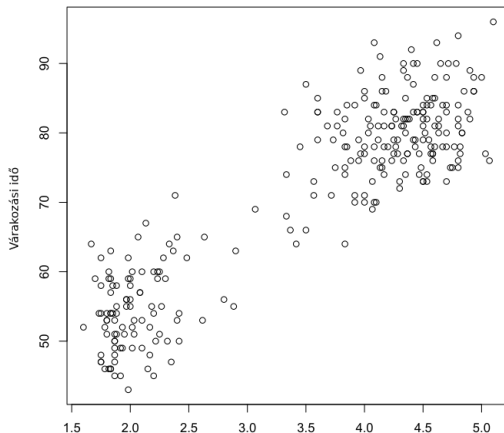
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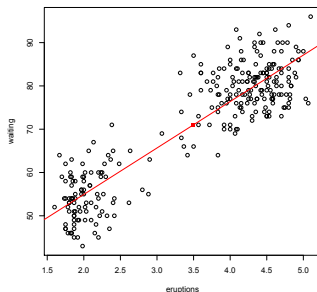
Linear Regression

Example (`faithful` dataset in R): the figure shows waiting times until a geyser eruption and the durations of the eruptions. The two variables exhibit positive correlation.



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Assume we know the joint distribution of the variables.

Goal: approximate the y -data by a **linear function** of the x -data $\rightarrow$ the „average” approximation error should be as small as possible.

Optimal line? $\rightarrow$ How do we measure the error?

Linear Regression

The next definition specifies how we measure the error and what we call the optimal linear function:

Definition

Let X and Y be random variables. The *linear regression* of Y on X is the random variable $\beta X + \alpha$, where $\alpha, \beta \in \mathbb{R}$, that minimizes

$$\mathbb{E}\left((Y - (\beta X + \alpha))^2\right).$$

Here:

$$\mathbb{E}\left(\underbrace{\left(\underbrace{Y}_{\text{„explained/dependent variable”}} - \underbrace{(\beta X + \alpha)}_{\substack{\text{„explanatory variable”} \\ \text{approximation (linear function)}}}\right)^2}_{\text{approximation error}}\right).$$

Linear Regression

Theorem

Let X and Y be random variables with $\mathbb{E}(X^2) < \infty$, $\mathbb{E}(Y^2) < \infty$, and $\mathbb{D}^2(X) \neq 0$. Then the expectation above is minimized exactly when

$$\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} \quad \text{and} \quad \alpha = \mathbb{E}(Y) - \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} \mathbb{E}(X).$$

(Proof: see lecture)

Definition

The regression line of Y on X is the line

$$\{(x, y) \in \mathbb{R}^2 : y = \beta x + \alpha\}$$

in the plane, where α and β are as in the proposition above.

Linear Regression: remarks

Reminder: correlation $\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\mathbb{D}X \mathbb{D}Y}$.

Claim:

$$\frac{(\beta X + \alpha) - \mathbb{E}(Y)}{\mathbb{D}(Y)} = \frac{X - \mathbb{E}(X)}{\mathbb{D}X} \text{corr}(X, Y).$$

In other words: if in the standardized version of $Y (= \frac{Y - \mathbb{E}(Y)}{\mathbb{D}(Y)})$ we replace Y by the linear regression $\beta X + \alpha$, then the result equals the correlation times the standardized X .

(This follows by a simple rearrangement.)

Linear Regression: example

$X \backslash Y$	0	1	p_Y
0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
2	0	$\frac{1}{4}$	$\frac{1}{4}$
p_X	$\frac{1}{2}$	$\frac{1}{2}$	1

We toss a fair coin twice, $X = \mathbb{1}_{\{\text{2nd toss is heads}\}}$, $Y =$ the total number of heads. We have: $\text{cov}(X, Y) = \frac{1}{4}$, $\mathbb{E}(X) = \frac{1}{2}$, $\mathbb{E}(Y) = 1$, $\mathbb{D}^2(X) = \frac{1}{4}$, $\mathbb{D}^2(Y) = \frac{1}{2}$. Hence

$$\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)} = 1, \quad \alpha = \mathbb{E}(Y) - \beta \mathbb{E}(X) = 1 - 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

The regression line is $\{(x, y) \in \mathbb{R}^2 : y = x + \frac{1}{2}\}$.

Variance of the Linear Regression Error

Theorem

Let the linear regression of Y on X be $\beta X + \alpha$. Then the variance of the deviation/error is

$$\mathbb{D}^2(Y - (\beta X + \alpha)) = \mathbb{D}^2(Y) - \frac{\text{cov}(X, Y)^2}{\mathbb{D}^2(X)}.$$

(Proof: by variance identities and using $\beta = \frac{\text{cov}(X, Y)}{\mathbb{D}^2(X)}$.)

Rearranging the equation:

$$\mathbb{D}^2(Y - (\beta X + \alpha)) = \mathbb{D}^2(Y) \left(1 - \text{corr}(X, Y)^2\right).$$

Thus the larger $|\text{corr}(X, Y)|$ is, the smaller the linear regression error. If $\text{corr}(X, Y) = \pm 1$, the error is 0, which is not surprising since in that case Y is a linear function of X , and by the definition of β, α we have $Y = \beta X + \alpha$.

Error of Linear Regression: example

Y \ X	X		p_Y
	0	1	
0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
2	0	$\frac{1}{4}$	$\frac{1}{4}$
p_X	$\frac{1}{2}$	$\frac{1}{2}$	1

We saw: $\text{cov}(X, Y) = \frac{1}{4}$, $\mathbb{D}^2(X) = \frac{1}{4}$, $\mathbb{D}^2(Y) = \frac{1}{2}$. Hence

$$\mathbb{D}^2(Y - (\beta X + \alpha)) = \mathbb{D}^2(Y) - \frac{\text{cov}(X, Y)^2}{\mathbb{D}^2(X)} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Of course, we can compute this directly as well ($\beta = 1$, $\alpha = \frac{1}{2}$):

$$\begin{aligned}\mathbb{D}^2(Y - (\beta X + \alpha)) &= \mathbb{D}^2(Y - (X + \tfrac{1}{2})) = \mathbb{D}^2(Y - X) \\ &= \mathbb{D}^2(Y) + \mathbb{D}^2(X) - 2\text{cov}(X, Y) = \frac{1}{2} + \frac{1}{4} - \frac{2}{4} = \frac{1}{4}.\end{aligned}$$

Conditional Expectation

(A difficult topic, sometimes with easy exam problems.)

Conditional Expectation: motivation

Reminder: conditional probability, $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ if $\mathbb{P}(B) > 0$.

Now: conditional expectation, starting with two questions.

Example 1 (discrete): We roll two dice. Given that the first roll equals $k \in [6]$, what is the expectation of the sum of the two rolls?

With our current toolkit, the „conditional expectation” in the question is not yet well-defined. Nevertheless, let us try to give an *intuitively correct* answer → motivation for the forthcoming definitions.

Conditional Expectation: introductory discrete example

Example 1 (discrete): We roll two dice. Given that the first roll equals $k \in [6]$, what is the expectation of the sum?

Answer:

Since the two rolls are independent, knowing that the first roll equals the constant k tells us nothing about the outcome of the second roll.

Therefore the **conditional expectation** is the sum of k and the expectation of a fair die roll, i.e., $k + 3.5$.

Note: it is not enough to appeal to linearity of expectation; we also need **independence**.

If the two rolls were not independent, the result need not be $k + 3.5$. For instance, if we forced the second roll to equal the first, then given the first is k , the expected sum would be $2k$.

Conditional Expectation: introductory discrete example

Example 1: Roll two dice. Let X be the first and Y the second outcome. Given $X = k \in [6]$, the expectation of $X + Y$ is $k + 3.5$.

Definition of conditional expectation that yields this result?

Conditional Expectation: introductory discrete example

Example 1: Roll two dice. Let X be the first and Y the second outcome. Given $X = k \in [6]$, the expectation of $X + Y$ is $k + 3.5$.

Definition of conditional expectation that yields this result?

We saw: $\mathcal{F} \rightarrow [0, 1]$, $A \mapsto \mathbb{P}(A|X = k)$ is a probability measure if $\mathbb{P}(X = k) > 0$.
For a random variable Z set

$$\mathbb{E}(Z|X = k) \stackrel{\text{def}}{=} \sum_{i \in \text{Ran}(Z)} i \mathbb{P}(Z = i|X = k), \quad k \in \text{Ran}(X).$$

Observe that for $l = 1, \dots, 6$ we have

$$\mathbb{P}(X + Y = k + l|X = k) = \mathbb{P}(Y = l|X = k) = \mathbb{P}(Y = l),$$

where the last step uses independence of X and Y . Hence

$$\begin{aligned} \mathbb{E}(X + Y|X = k) &= \sum_{i \in \text{Ran}(X+Y)} i \mathbb{P}(X + Y = i|X = k) \\ &= \sum_{l \in \text{Ran}(Y)} (l + k) \mathbb{P}(X + Y = l + k|X = k) = \sum_{l \in \text{Ran}(Y)} (l + k) \mathbb{P}(Y = l) \\ &= \sum_{l \in \text{Ran}(Y)} l \mathbb{P}(Y = l) + k \sum_{l \in \text{Ran}(Y)} \mathbb{P}(Y = l) = \mathbb{E}(Y) + k = 3.5 + k. \end{aligned}$$

Conditional Expectation: discrete case

We now formulate the example and the definitions in the general discrete setting.

Definition

Let Y be a random variable and A an event with $\mathbb{P}(A) > 0$. The *conditional expectation* of Y given A is the expectation of Y with respect to the probability measure $\mathbb{P}(\cdot|A)$. Notation: $\mathbb{E}(Y|A)$.

Interpretation: $\mathbb{E}(Y|A)$ is the average of Y restricted to the event A , instead of the whole sample space.

Lemma

Let Y be a simple random variable and A an event with $\mathbb{P}(A) > 0$. Then

$$\mathbb{E}(Y|A) = \sum_{k \in \text{Ran}(Y)} k \mathbb{P}(Y = k|A).$$

(The proof is analogous to the formula for $\mathbb{E}(Y)$.)

Conditional Expectation: discrete case

Reducing conditional expectation to the expectation formula:

Theorem

Let Y be a random variable and suppose $\mathbb{P}(A) > 0$. Then

$$\mathbb{E}(Y|A) = \frac{\mathbb{E}(Y \cdot \mathbb{1}_A)}{\mathbb{P}(A)}.$$

Here $Y \cdot \mathbb{1}_A$ is defined by pointwise multiplication of random variables:

$$(Y \cdot \mathbb{1}_A)(\omega) = Y(\omega)\mathbb{1}_A(\omega) = \begin{cases} Y(\omega), & \text{if } \omega \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Example

Let X and Y again be two independent die rolls.
Compute $\mathbb{P}(Y = k | X + Y < 7)$ for $k = 1, \dots, 6$.

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$$\begin{aligned}\mathbb{P}(Y = k | X + Y < 7) &= \frac{\mathbb{P}(Y = k, X + Y < 7)}{\mathbb{P}(X + Y < 7)} = \frac{\mathbb{P}(Y = k, X < 7 - k)}{\frac{15}{36}} \\ &= \frac{\mathbb{P}(Y = k)\mathbb{P}(X < 7 - k)}{\frac{15}{36}} = \frac{\frac{1}{6} \cdot \frac{7-k-1}{6}}{\frac{15}{36}} = \frac{6-k}{15}.\end{aligned}$$

Therefore

$$\begin{aligned}\mathbb{E}(Y | X + Y < 7) &= \sum_{k=1}^6 k \mathbb{P}(Y = k | X + Y < 7) = \sum_{k=1}^6 k \frac{6-k}{15} \\ &= \frac{6}{15}(1 + \dots + 6) - \frac{1}{15}(1 + 4 + \dots + 36) = \frac{6 \cdot 21}{15} - \frac{91}{15} = \frac{35}{15} = \frac{7}{3}.\end{aligned}$$

Example, continued

Since $X + Y$ is discrete, events of the form $\{X + Y = k\}$ can have positive probability. Thus $\mathbb{E}(Y|X + Y = 5)$ is meaningful, and

$$\begin{aligned}\mathbb{E}(Y|X + Y = 5) &= \sum_{k=1}^6 k\mathbb{P}(Y = k|X + Y = 5) \\&= \sum_{k=1}^6 k \frac{\mathbb{P}(Y = k, X + Y = 5)}{\mathbb{P}(X + Y = 5)} \\&= \sum_{k=1}^6 k \frac{\mathbb{P}(Y = k)\mathbb{P}(X = 5 - k)}{\mathbb{P}(X + Y = 5)} \\&= \sum_{k=1}^4 k \frac{\frac{1}{6} \cdot \frac{1}{6}}{\frac{4}{36}} = \frac{10}{4} = 2.5.\end{aligned}$$

This is unsurprising: conditioned on $X + Y = 5$, X is on average 2.5, since X and Y are i.i.d. random variables.

Conditional Expectation: discrete case

Returning to the example: now that we know

$$\mathbb{E}(X + Y|X = k) = k + 3.5, \quad k = 1, \dots, 6,$$

let us ask something stronger.

What is the expectation of $X + Y$, **given X** (notation: $\mathbb{E}(X + Y|X)$)?

Conditional Expectation: discrete case

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What is the expectation of $X + Y$, **given X** (notation: $\mathbb{E}(X + Y|X)$)?

Since X is random, the answer is not a number. It will be a **random variable**, „the best approximation to $X + Y$ given X .“

Given $X = k$, the conditional expectation of $X + Y$ is $k + 3.5$.

Thus the answer is $\mathbb{E}(X + Y|X) = X + 3.5$.

Definition

Conditional expectation in the discrete case Let X and Y be random variables with X discrete. Let S_X be the set of essential values of X ,

$$S_X \stackrel{\text{def.}}{=} \{x \in \mathbb{R} \mid \mathbb{P}(X = x) > 0\}.$$

Define the function $g: S_X \rightarrow \mathbb{R}$, $g(x) = \mathbb{E}(Y|X = x)$. Then the *regression* (=conditional expectation) of Y on X is the random variable $g(X)$. Notation: $\mathbb{E}(Y|X)$.

Conditional Expectation: another discrete example

Example: Roll a fair die, then flip as many fair coins as the die shows. Let X be the die result, and Y the number of heads among the coin flips. What is the regression of Y on X ?

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Given the value of X , the distribution of Y is binomial with parameters X and $1/2$.

Hence $\mathbb{E}(Y|X = x) = x \cdot \frac{1}{2} =: g(x)$, i.e., $\mathbb{E}(Y|X) = g(X) = \frac{X}{2}$.

In general: **regression = conditional expectation (of one random variable given another) $\neq$ linear regression.**

Linear regression: best approximation of Y by a **linear** function of X .

Regression: does not restrict to linear functions; it returns the function $g(X)$ that „best approximates“ Y given X . In what sense? $\rightarrow$ next slide.

Regression as best approximation

Informally, the random variable $\mathbb{E}(Y|X)$ „captures everything about Y that can be known from X ,“ and provides an approximation to Y with the „maximum“ available information.

More precisely: if we impose an additional condition involving X and look at the conditional expectation of Y under that condition, it can be computed from $\mathbb{E}(Y|X)$ alone, without access to the original Y .

Theorem

Let X be a discrete random variable. Assume $\mathbb{P}(X < x) > 0$ and $\mathbb{E}(|Y|) < \infty$. Then

$$\mathbb{E}[g(X)|X < x] = \mathbb{E}(Y|X < x)$$

for all $x \in \mathbb{R}$, where $g(X) = \mathbb{E}(Y|X)$.

(Proof: see lecture.)

Regression in the continuous case: first example

Example 2 (continuous): We simulate $X \sim U(0, 1)$ on a computer. We are told that the result exceeds $1/2$. Given this, what is the expectation?

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Heuristic answer: since X is uniform, its density is constant 1 on $(0, 1)$. Conditioning on $X > 1/2$ yields a constant density on $(1/2, 1)$ (now constant 2 to integrate to 1). Thus $\mathbb{E}(X|X > 1/2)$ is the midpoint of the interval, i.e., $3/4$.

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Proper answer: since $\mathbb{P}(X > 1/2) = \frac{1}{2} > 0$, by the previous definition

$$\begin{aligned}\mathbb{E}(X|X > 1/2) &= \frac{\mathbb{E}(X \cdot \mathbb{1}_{\{X \in (1/2, 1)\}})}{\mathbb{P}(X > 1/2)} = \frac{\int_{1/2}^1 x f_X(x) dx}{1/2} \\ &= \frac{\int_{1/2}^1 x dx}{1/2} = \frac{1/2 - 1/8}{1/2} = \frac{3}{4}.\end{aligned}$$

Regression in the continuous case

Assume now that X is continuous. For a given Y we want to define $\mathbb{E}(Y|X)$ as the „best approximation to Y given X ,“ analogous to the discrete case.

For a discrete Z we had:

$$\mathbb{E}[Y|Z = z] = \sum_{k \in \text{Ran}(Y)} k \mathbb{P}(Y = k|Z = z),$$

provided $\mathbb{P}(Z = z) > 0$.

Problem: for continuous X we have $\mathbb{P}(X = x) = 0$ for all $x \in \mathbb{R}$. Hence the formula above and classical conditional probabilities do **not** define $\mathbb{E}(Y|X = x)$!

This motivated the **conditional density** when (X, Y) is a continuous random vector:

$$f_{Y|X}(y|x) = \frac{f_{(X,Y)}(x,y)}{f_X(x)},$$

if $f_X(x) > 0$, and 0 otherwise.

Conditional density and regression

Conditional density:

$$f_{Y|X}(y|x) = \frac{f_{(X,Y)}(x,y)}{f_X(x)},$$

if $f_X(x) > 0$, and 0 otherwise.

Earlier (informally) we said this behaves like $f_Y(y)$, but conditional on X being near x .

We know $\mathbb{E}(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$. Thus we expect that

$$\mathbb{E}(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

is a good definition. (This is again **formal** notation, not classical conditional probability.)

Introduce $g(x) = \mathbb{E}(Y|X = x)$. In the discrete case we had $\mathbb{E}(Y|X) = g(X)$.

Question: Is the same true more generally (e.g., for continuous X)? What is the definition of regression in that case?

